

RUT Prediction Note: Wave-Based Test Lanes

GitHub / OSF v1 - gravitational-wave H0, neutron-star phase signatures, and cross-channel ripple consistency

Purpose

This note converts today's wave, quantum-gravity, and phase-state inserts into a compact prediction document. It is designed for GitHub, OSF, and outreach packets, not as a claim that recent articles prove Reverse Universe Theory. The goal is narrower and stronger: define new test lanes where RUT's omega, ripple, and phase-ladder sectors can be compared against future data.

Reviewer-safe framing: These predictions are not declarations of discovery. They are proposed falsifiable targets. RUT gains credibility only if the same compact sector survives across independent wave-based, expansion-based, and structure-based probes.

Prediction 1 - Gravitational-wave H0 as an omega-sector test

If RUT's omega/phase-ripple sector is physically meaningful, gravitational-wave expansion measurements should become one of the cleanest independent constraints on the theory. Future standard-siren and stochastic-siren measurements should be compared against the same parameters used in electromagnetic distance-ladder, BAO, and CMB-facing fits.

Operational statement:

$H0_{GW}(\theta_{\omega})$ compared with $H0_{EM}(\theta_{\omega})$, $H0_{BAO}(\theta_{\omega})$,
 $H0_{CMB}(\theta_{\omega})$

Falsifier: If gravitational-wave H0 measurements converge tightly on a standard LambdaCDM-compatible value while RUT requires a substantially different value from the same omega/ripple parameterization, that active parameterization is constrained or rejected.

Positive target: If gravitational-wave H0 measurements reveal a consistent high-H0, epoch-dependent, or scale-dependent expansion signal compatible with the same RUT parameter region used in BAO/ripple fits, the omega/phase-ripple sector becomes more interesting.

Prediction 2 - Neutron-star phase-ladder waveform signature

If RUT's phase-ladder idea has physical value, extreme matter transitions inside neutron stars should not appear only as equation-of-state changes. They should leave measurable waveform fingerprints: mode shifts, bandwidth changes, damping differences, or post-merger ringing features in gravitational-wave data.

Operational statement:

$\Delta\omega / \omega_{\text{center}} = 1 - \exp[-\alpha S / (N k_B)]$

In a cautious RUT reading, the entropy-bandwidth law becomes a candidate diagnostic: phase transitions and coherence loss should widen, shift, or damp oscillation modes in a way that can be compared against neutron-star merger spectra.

Falsifier: If realistic neutron-star waveform modeling shows no distinct RUT-linked bandwidth, mode, or phase-transition signature beyond standard equation-of-state degeneracy, then the phase-ladder claim remains analogy rather than physics.

Prediction 3 - Cross-channel ripple consistency

A serious RUT omega/ripple sector should not win in one observable lane while failing everywhere else. The same compact parameter region should be tested across BAO/ripple horizon behavior, gravitational-wave expansion inference, growth/lensing structure, and early-universe relic constraints.

Operational statement:

θ_{ω}^* must remain viable across {BAO/ r_d , GW-H0, growth/lensing, CMB/early relics}

Falsifier: If every successful RUT fit requires a different omega/ripple parameter region for each probe, the framework becomes too flexible and loses explanatory force.

Positive target: If one shared parameter region improves or remains viable across multiple independent channels, RUT becomes a more serious multi-probe research program rather than a single-fit anomaly.

Recommended next step

Upload this note with the v28 books and the Penrose Packet, then move to the next simulation. The most valuable next simulation is one that tests transfer: whether the same omega/ripple structure that motivates the enlarged horizon can remain consistent with a second probe lane, especially gravitational-wave H_0 , growth/lensing, or a cleaner r_d closure.

Suggested filename

RUT_Prediction_Note_Wave-Based_Test_Lanes_v1.pdf

Source notes

Space.com, Robert Lea, "What is quantum gravity? Scientists think it could explain the beginning of our universe," published April 29, 2026.

Space.com, Robert Lea, "How fast is the universe actually expanding? Ripples in spacetime could finally solve Hubble tension," published March 2, 2026.

Space.com, Keith Cooper, "A state of matter last seen just after the Big Bang may exist inside neutron stars - and scientists think they can prove it," published March 15, 2026.

Use note: These sources motivate test lanes. They are not treated as proof of RUT.